

# Christopher Musco

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## Academic Positions

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| <b>New York University</b><br><i>Assistant Professor, Tandon School of Engineering</i><br><i>Department of Computer Science and Engineering</i> | <b>New York, NY</b><br><i>2019 - present</i> |
| <b>Princeton University</b><br><i>Research Instructor, Department of Computer Science</i>                                                       | <b>Princeton, NJ</b><br><i>2018 – 2019</i>   |
| <b>Massachusetts Institute of Technology</b><br><i>Research and Teaching Assistant, Theory of Computation Group</i>                             | <b>Cambridge, MA</b><br><i>2013 – 2018</i>   |
| <b>Yahoo Labs</b><br><i>Research Intern, Scalable Machine Learning Group</i>                                                                    | <b>New York, NY</b><br><i>Summer 2015</i>    |

## Research Areas

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foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms  
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

## Education

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| <b>Massachusetts Institute of Technology</b><br><i>Ph.D. in Computer Science</i><br>Advisor: Jonathan Kelner<br>Thesis: Faster Linear Algebra for Data Analysis and Machine Learning<br>M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices | <b>Cambridge, MA</b><br><i>2013 – 2018</i> |
| <b>Yale University</b><br><i>B.S. in Applied Mathematics, B.S. in Computer Science</i>                                                                                                                                                                           | <b>New Haven, CT</b><br><i>2008 – 2012</i> |

## Publications

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*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a \*, which indicates that authors are ordered by contribution.*

**Graph Learning for Inverse Landscape Genetics.** Prathamesh Dharangutte, Christopher Musco. *In submission.* 2020.

**Fourier Sparse Leverage Scores and Approximate Kernel Learning.** Tamás Erdélyi, Cameron Musco, Christopher Musco. *In submission.* 2020.

**Near Optimal Linear Algebra in the Online and Sliding Window Models.** Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *In submission.* 2020.

**The Statistical Cost of Robust Kernel Hyperparameter Turning.** Raphael A. Meyer, Christopher Musco. *In submission.* 2020.

**Towards Equitable Transit Planning\***. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *In submission*. 2020.

**Finding the Mode of a Kernel Density Estimate**. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *In submission*. 2020.

**Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers**. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

**Sample Efficient Toeplitz Covariance Estimation**. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

**Fast and Space Efficient Spectral Sparsification in Dynamic Streams**. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

**Understanding Filter Bubbles and Polarization in Social Networks**. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM)* 2020.

**Low-Rank Approximation from Communication Complexity**. Cameron Musco, Christopher Musco, David Woodruff. *In submission*. 2019.

**A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms**. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

**Learning Networks from Random Walk-Based Node Similarities**. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

**Eigenvector Computation and Community Detection in Asynchronous Gossip Models**. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

**Minimizing Polarization and Disagreement in Social Networks**. Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

**Stability of the Lanczos Method for Matrix Function Approximation**. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

**Recursive Sampling for the Nyström Method**. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

**Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees**. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

**Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling**. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

**Determining Tournament Payout Structures for Daily Fantasy Sports**. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

**Principal Component Projection Without Principal Component Analysis**. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

**Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition**. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

**Dimensionality Reduction for K-Means Clustering and Low Rank Approximation**. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of*

*Computing (STOC) 2015.*

**Principled Sampling for Anomaly Detection.** Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS) 2015.*

**Uniform Sampling for Matrix Approximation.** Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS) 2015.*

**Single Pass Spectral Sparsification in Dynamic Streams.** Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS) 2014, Special Issue SIAM Journal on Computing 2017.*

## Student Research Advising

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|----------------------------------------------------|----------------|
| Apoorv Singh. Ph.D., New York University.          | 2020 - present |
| Raphael Meyer. Ph.D., New York University.         | 2019 - present |
| Prathamesh Dharangutte. M.S., New York University. | 2019 - present |
| John Shin. M.S., New York University.              | 2019 - present |
| Tristan Pollner. Undergraduate, Princeton.         | 2018 - 2019    |

## Teaching

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### **Introduction to Machine Learning (NYU CS-UY 4563)**

*Course Instructor* Spring 2020

### **Algorithmic Machine Learning and Data Science (NYU CS-GY 9223I)**

*Course Instructor* Fall 2019

### **Advanced Algorithm Design (Princeton COS 521)**

*Course Instructor* Fall 2018

### **Advanced Algorithms (MIT 6.854/18.415)**

*Teaching Assistant* Spring 2016

### **Technical Communication Skills for Graduate Students (MIT 6.S977)**

*Workshop Leader* Spring 2016

### **Introduction to Computer Science (Yale CPSC-201)**

*Teaching Assistant* Spring 2012

### **Introduction to Programming (Yale CPSC-112)**

*Teaching Assistant* Fall 2011

## Mentorship and Outreach

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### **SIAM/MathWorks Math Modeling Challenge**

*Problem Development Committee, Judge* 2018 – present

Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Score and provide feedback on student submissions to the contest.

### **MIT Graduate Communication Lab**

*Communication Adviser* 2015 – 2018

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

## CSAIL Algorithms Office Hours

Member

2016 – 2018

Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

## Research Science Institute at MIT

Summer Research Adviser

2014

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

## Talks and Presentations

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### Introduction to Machine Learning

New York University, guest lecture in "Introduction to Computer Science"

Mar. 2020

### Structured Covariance Estimation

Yahoo! Research, Seminar

April 2020

Johns Hopkins University, Theory Seminar

Oct. 2019

Symposium on Discrete Algorithms (SODA)

Jan. 2020

### From Randomized Linear Algebra to Randomized Functional Analysis

SIAM Mathematics of Data Science Conference

May 2020

Rutgers University, DIMACS RandNLA Workshop

Sept. 2019

University of Utah, Data Science Seminar

Jan. 2020

### Learning Signals with Simple Fourier transforms

Symposium on Theory of Computing (STOC)

June 2019

Columbia University, Theory Seminar

Oct. 2018

UC Berkeley, Simons Institute Workshop

Sept. 2018

Princeton University, Theory Seminar

Sept. 2018

### The Lanczos Method in Data Science: Challenges and Opportunities

SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium

Feb. 2020

Georgetown University, CS Seminar

Feb. 2018

Symposium on Discrete Algorithms (SODA)

Jan. 2018

### Unifying Random Features and Leverage Scores for Kernel Matrix Approximation

Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar

Feb. 2018

Harvard University, guest lecture in "Sketching Algorithms for Big Data"

Nov. 2017

Microsoft Research, Redmond

Sept. 2017

### Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge

July 2017

### Recursive Sampling for the Nyström Method

Neural Information Processing Systems (NeurIPS) poster session

Dec. 2017

New England Machine Learning Day poster session

May 2017

### Algorithms for Determining Tournament Payout Structures

Meeting on Algorithm Engineering and Experiments (ALENEX)

Jan. 2017

### Introduction to Linear Sketching

MIT, Theory Retreat

Sept. 2016

### Principal Component Projection without Principal Component Analysis

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|----------------------------------------------------------------|------------------|
| International Conference on Machine Learning (ICML)            | <i>June 2016</i> |
| <b>Ridge Leverage Score Sampling</b>                           |                  |
| National Institute of Informatics, Shonan Meeting              | <i>July 2016</i> |
| MIT, Algorithms & Complexity Seminar                           | <i>July 2016</i> |
| University of Utah, Data Group Meeting                         | <i>Jan. 2016</i> |
| <b>Randomized Block Krylov Methods</b>                         |                  |
| Neural Information Processing Systems (NeurIPS) poster session | <i>Dec. 2015</i> |
| <b>Large Scale Column Subset Selection</b>                     |                  |
| Yahoo Labs, Science Week poster session                        | <i>July 2015</i> |
| <b>Dimensionality Reduction for k-Means Clustering</b>         |                  |
| IBM T.J. Watson, Mathematical Sciences research seminar        | <i>Aug. 2015</i> |
| Symposium on Theory of Computing (STOC)                        | <i>June 2015</i> |
| MIT, Theory Lunch                                              | <i>June 2014</i> |
| <b>Principled Sampling for Anomaly Detection</b>               |                  |
| MIT, Defense Advanced Research Projects Agency site visit      | <i>Mar. 2015</i> |
| Network and Distributed System Security Symposium (NDSS)       | <i>Feb. 2015</i> |
| <b>Uniform Sampling for Matrix Approximation</b>               |                  |
| Yahoo Labs, Scalable Machine Learning research seminar         | <i>June 2015</i> |
| <b>Single Pass Spectral Sparsification in Dynamic Streams</b>  |                  |
| MIT, Annual Sublinear Algorithms Day poster session            | <i>Apr. 2015</i> |
| Harvard University, Theory Seminar                             | <i>Nov. 2014</i> |
| Foundations of Computer Science (FOCS)                         | <i>Oct. 2014</i> |

## Service

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| Conference Programming.....                                                                                                                                                                                                                                                                                                                                                                                   |                       |
| NeurIPS Area Chair                                                                                                                                                                                                                                                                                                                                                                                            | <i>2019</i>           |
| Organization.....                                                                                                                                                                                                                                                                                                                                                                                             |                       |
| SIAM Mathematics of Data Science Minisymposium Organizer                                                                                                                                                                                                                                                                                                                                                      | <i>2020</i>           |
| NYU Tandon Algorithmic Data Science Reading Group                                                                                                                                                                                                                                                                                                                                                             | <i>2019 - present</i> |
| Department Service.....                                                                                                                                                                                                                                                                                                                                                                                       |                       |
| Faculty Search Committee                                                                                                                                                                                                                                                                                                                                                                                      | <i>2019</i>           |
| Ph.D. Admissions Committee                                                                                                                                                                                                                                                                                                                                                                                    | <i>2019 - present</i> |
| Conference Reviewing.....                                                                                                                                                                                                                                                                                                                                                                                     |                       |
| STOC, FOCS, SODA, ITCS, ICALP, ESA, ICML, NeuRIPS, APPROX, RANDOM, STACS, IWOC, WALCOM, NDSS                                                                                                                                                                                                                                                                                                                  |                       |
| Journal Reviewing.....                                                                                                                                                                                                                                                                                                                                                                                        |                       |
| ACM Transactions on Algorithms, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science. |                       |

Grants Panels.....  
NSF Division of Information and Intelligent Systems (2019)  
Notables.....  
ICML 2018 Outstanding Reviewer

## Other Research Experience

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### **Rutgers, DIMACS**

*Invited participant* 2019  
Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.

### **UC Berkeley, Simons Institute for the Theory of Computing**

*Invited participant* 2018  
Workshop on Randomized Numerical Linear Algebra and Applications.

### **Japan National Institute of Informatics**

*Invited participant* 2016  
Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

## Honors and Awards

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### **National Science Foundation**

2014-2017

*Graduate Research Fellowship, recipient.*

### **Yale University**

2012

*Cum Laude, Distinction in Both Majors.*

## Other Professional Experience

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### **Redfin**

**Seattle, WA**

*Software Engineer, Data Team*

2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

### **Elysium Digital**

**Cambridge, MA**

*Summer Consultant*

2011

Provided technical expertise in software patent cases at leading litigation consulting company.

### **Amicus**

**New Haven, CT**

*Software Developer*

2010 – 2011

Built applications for political campaign management and social fundraising at an early-stage startup that went on to raise 4 million dollars in seed funding.